

The Caffeine Crew

Project Technical Appendix and Analyses Files

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1. Data Description

1.1 Data Collection

Our data was collected from Kaggle at the following link:

“<https://www.kaggle.com/datasets/ahmedabbas757/coffee-sales/data>”. The dataset is titled Coffee Shop Sales and contains synthetic data on three fictional coffee shops in New York. The dataset was originally uploaded by Ahmed Abbas, who made the data open to the public for analysis and educational use through the Kaggle platform.

The dataset is shared under the GNU Lesser General Public License (LGPL), which allows users to download, modify, and analyze the data for educational and non-commercial purposes as long as proper attribution is maintained. The LGPL states “Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.” Because of this licensing, no special permissions, data agreements, or formal approval were required for our group to use the dataset for academic research as part of this course. This license allows users to download, analyze, transform, and build projects from the dataset, as long as the original author is credited and the data is not misrepresented as our own original creation.

1.2 Variables

This dataset included 11 variables and 149,116 rows. Each row was a separate transaction at a particular store.

List of Variables

- transaction_id : Unique sequential ID representing an individual transaction. (int)
- transaction_date : Date of the transaction (MM/DD/YY) (dateTime)
- transaction_time : Timestamp of the transaction (HH:MM:SS) (object)
- transaction_qty : Quantity of items sold (int)
- store_id : Unique ID of the coffee shop where the transaction took place (int)
- store_location : Location of the coffee shop where the transaction took place (object)
- product_id : Unique ID of the product sold (int)
- unit_price : Retail price of the product sold (float)
- product_category : Description of the product category (object)
- product_type : Description of the product type (object)
- product_detail : Description of the product detail (object)

2. Data Processing

We used Python for all of our data processing, modeling, and visualizations. Some of the packages we used were “statsmodels.api”, “pandas”, “matplotlib.pyplot”, “seaborn”, and “matplotlib.ticker”.

We didn’t have to do much to clean the original dataset, because it was already in good condition. However, we did have to transform some of the data to do our analysis.

For example, to do our regressions, we created a new variable, “Revenue” by multiplying “transaction_qty” by “unit_price”. This allowed us to see the specific revenue that each sale brought.

In order to look at what day of the week had the most revenue, we had to convert the transaction type to day of week, and then get dummy variables for each one. We did the same for each month also. Then we had to convert these variables from booleans into integers to run the regression.

For the second regression, we derived the transaction hour using a lambda function. Then we binned each transaction into three categories; morning(5:00-12:00 pm), afternoon(12:00-5:00 pm) and evening(5:00pm-10:00 pm). We created dummy variables for this and dummy variables for each product_category. There were 9 total product categories.

3. Analytical Methods/Visualizations

All analyses were performed in Python using pandas for data manipulation, matplotlib/seaborn for visualizations, and statsmodels for regression modeling. Each method used in the project directly supports the marketing question: *How can a coffee shop optimize the timing and type of promotions to increase customer engagement and sales?*

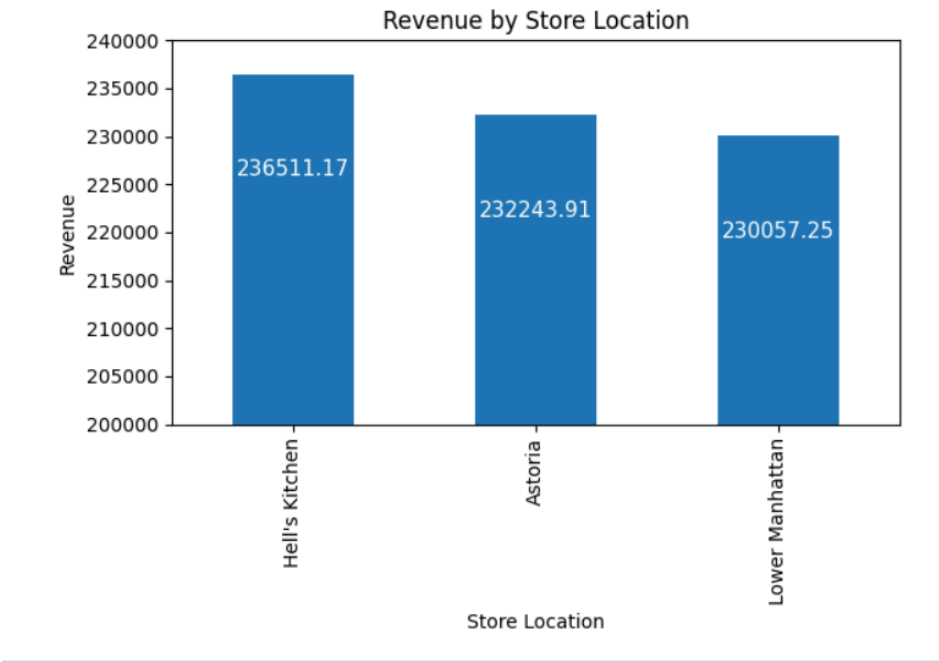
3.1 Time-Series Trend Analysis

To examine how revenue evolved throughout 2023, we aggregated the total revenue for each calendar day. This allowed us to construct a continuous time series that displayed overall business performance across the year. We then applied a time-series regression model where daily revenue was regressed against a linear time index and dummy variables representing each day of the week. This model helped identify both long-term trends and recurring weekly cycles.

The results revealed a strong, statistically significant upward trend, showing that revenue steadily increased from approximately \$2,000–\$3,000 per day in early 2023 to about \$5,000–\$6,000 per day by the year’s end. The regression also highlighted consistent weekly patterns: Mondays were the strongest revenue days, while Tuesdays were the weakest, a pattern repeated across many

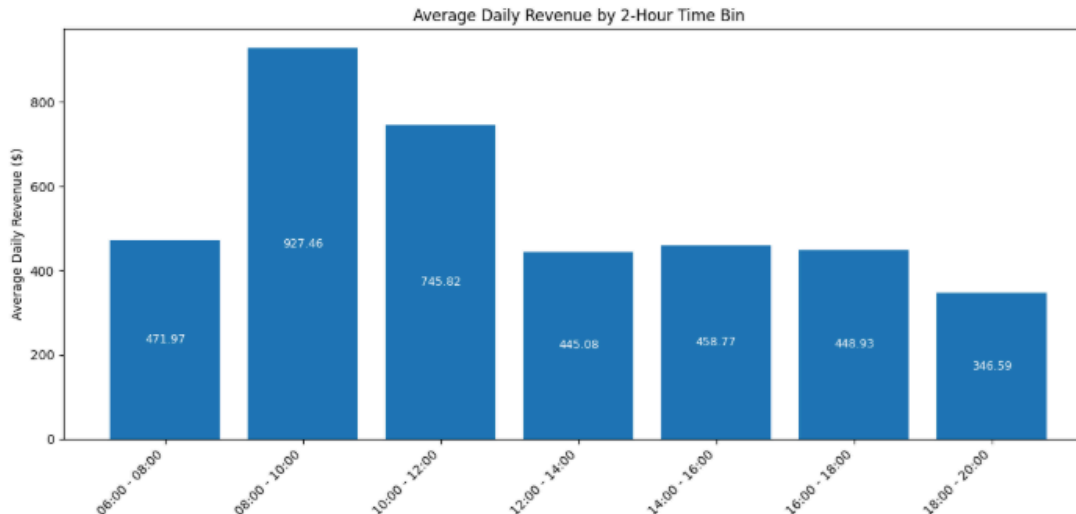
months. Seasonal effects also emerged, with higher-than-average revenue observed during late spring and early summer (May–June). Collectively, these findings confirm both strong business growth and stable weekly demand cycles, which are crucial for promotion planning, staffing decisions, and inventory optimization.

3.2 Revenue by Store Location



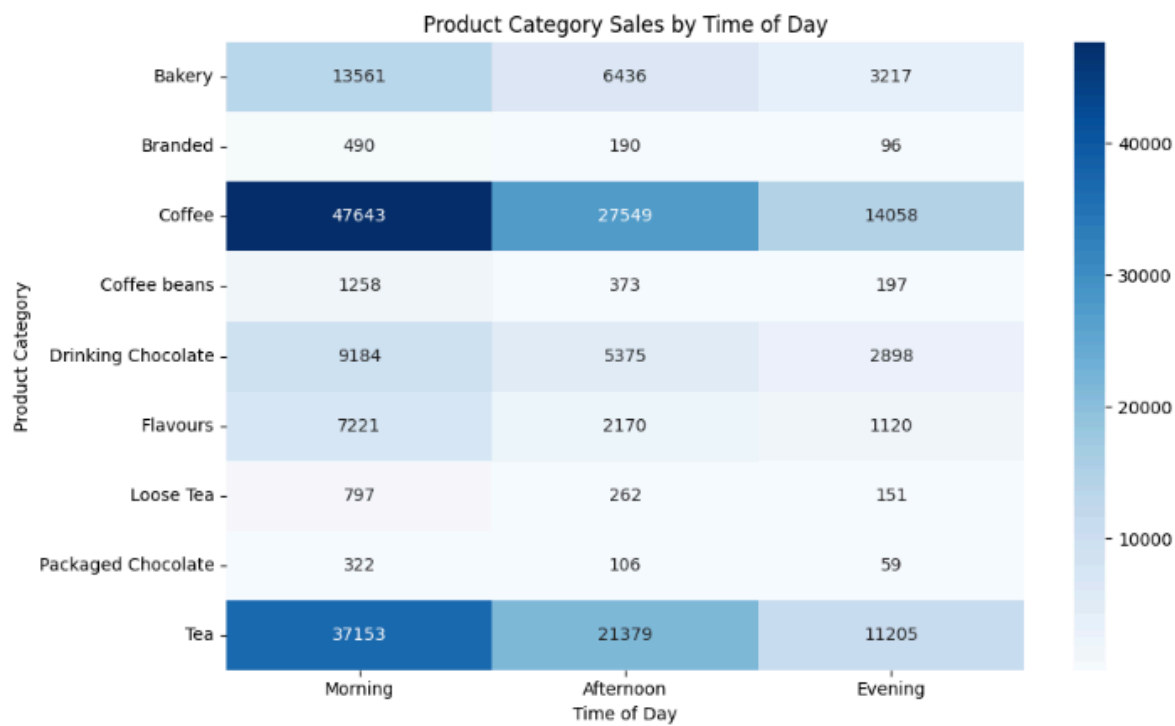
We examined store-level performance by aggregating total annual revenue for each of the three NYC locations. The analysis showed that Hell’s Kitchen generated the highest yearly revenue (\$236,511), closely followed by Lower Manhattan (\$236,057) and Astoria (\$232,243). Despite slight differences, all three stores performed remarkably similarly, indicating a well-balanced customer base across the city. This suggests that the brand has strong geographic consistency, which supports scalable promotion strategies. The slight lead of Hell’s Kitchen and Lower Manhattan also suggests that these locations may benefit from expanded morning promotions, additional barista staffing, or inventory prioritization.

3.3 Peak Hours Analysis



To identify the busiest and slowest times of day, we computed average revenue for each hour across all days in 2023. The results were clear and consistent: the morning hours between 7 AM and 12 {< generated the highest revenue, averaging around \$927. This reflects strong breakfast and early commute demand. Revenue during the afternoon period (12 PM to 5 PM) stayed relatively stable at approximately \$450, while revenue dropped sharply after 6 PM, averaging around \$346. These patterns highlight distinct demand-driven time segments and have direct managerial implications, particularly for staffing and promotional scheduling. For example, mornings should have the highest staffing levels, while afternoon and evening sales can be boosted through targeted promotions, loyalty rewards, or product bundling.

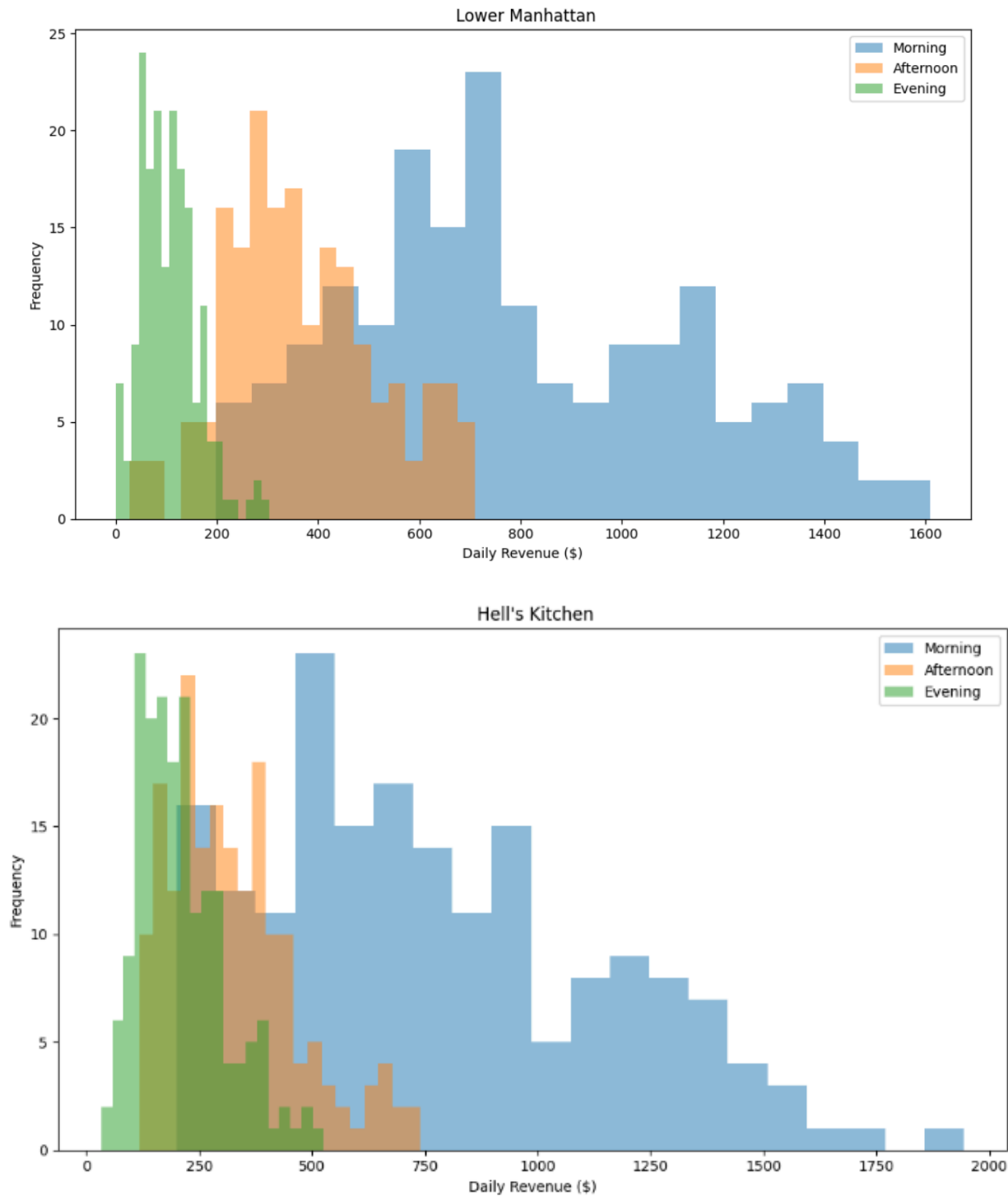
3.4 Product Category Heat Map

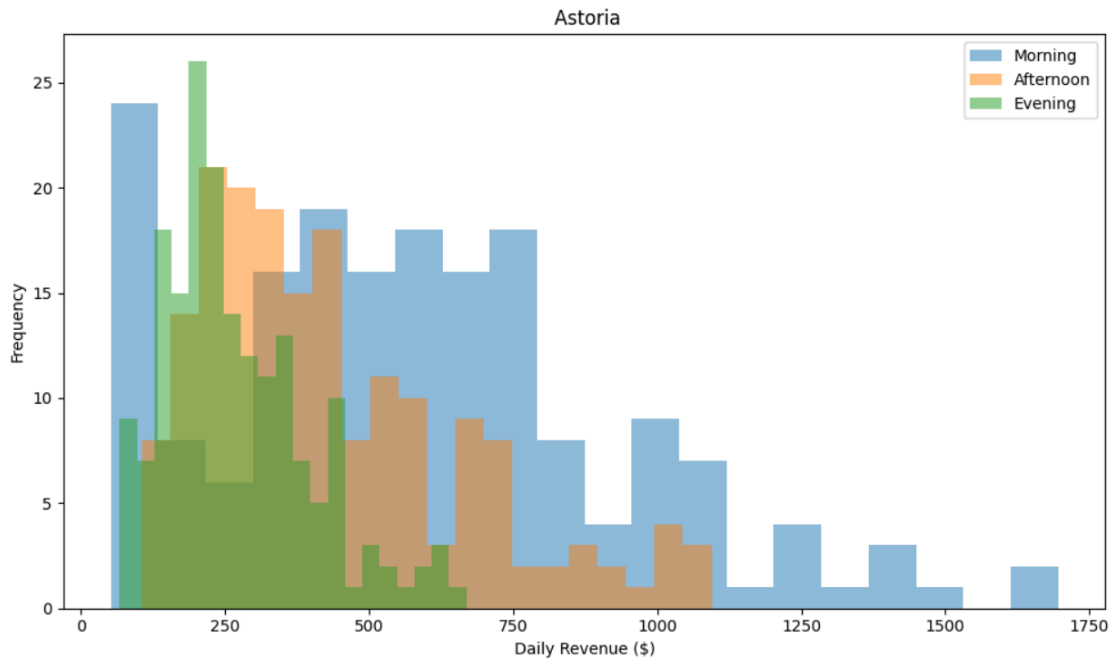


To understand how product popularity varied throughout the day, we built a revenue heat map that cross-tabulated product categories with each time period of the day. This visualization revealed that the morning period overwhelmingly dominated total sales across nearly every category. Coffee and tea had especially strong morning demand, consistent with typical café consumption habits. Bakery items, drinking chocolates, and even packaged goods saw their highest activity in the morning and early afternoon. As the day progressed, sales declined gradually, and evenings showed notably lower performance across almost all categories. This heat map provided crucial insights about category-specific scheduling and promotional

timing—for instance, highlighting which items should be emphasized during morning rush hours versus which categories might need promotional support later in the day.

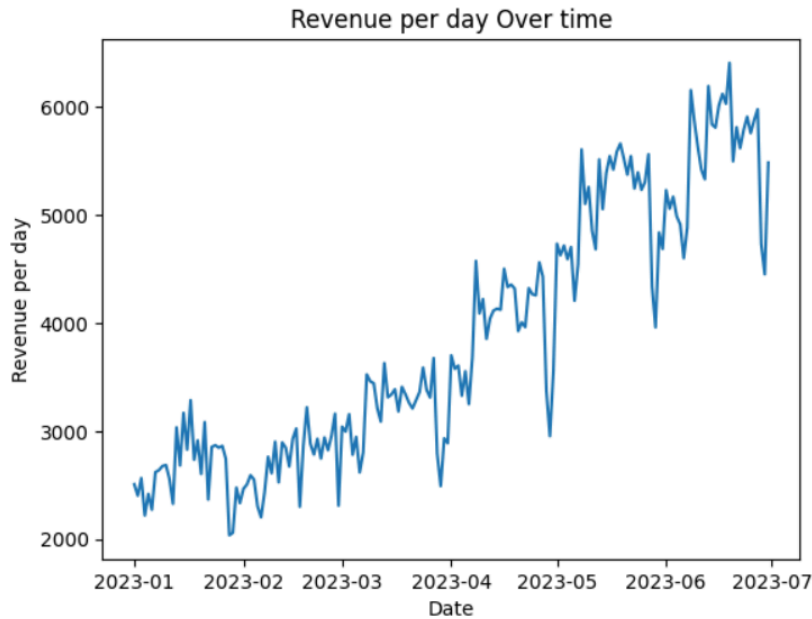
3.5 Revenue Layered Histograms





To further understand differences between store performance across time-of-day segments, we created layered histograms that compared daily revenue distributions during morning, afternoon, and evening time blocks across all three locations. This analysis revealed that Astoria consistently outperformed the other two stores in both the morning and afternoon periods, suggesting unique local demand characteristics or stronger product mix effectiveness in that neighborhood. Conversely, Hell’s Kitchen underperformed during the afternoon, indicating potential operational or competitive challenges during that time window. These results offer valuable, store-specific strategic insights—such as identifying which store may require targeted afternoon marketing, operational adjustments, or product changes. The Lower Manhattan store underperformed during the morning, as compared to the other two. It is important for the company to study each store, get to know their differences, and try to understand why some stores are performing better at different times of the day. Once they see why a store is performing well at that time, they can try and replicate this in the other stores.

3.6 Revenue over Time



There was a clear upward trend in our revenue per day over the time. The first two months seemed a little stagnant, but then afterwards revenue showed growth every month. It is also interesting to note that revenue seemed to hit a drastic dip at around the same time every month. To further analyze this, we calculated day over day revenue change, and then made a list in descending order showing the 10 days where the shops lost the most revenue.

	transaction_date	Revenue	Rev_Change
178	2023-06-28	4728.90	-1246.75
147	2023-05-28	4338.65	-1220.50
117	2023-04-28	3373.80	-1053.30
170	2023-06-20	5494.66	-909.25
86	2023-03-28	2792.55	-881.80
58	2023-02-28	2311.10	-848.90
47	2023-02-17	2300.75	-722.58
21	2023-01-22	2367.33	-715.52
27	2023-01-28	2037.10	-705.00
17	2023-01-18	2735.96	-549.84

Out of the 10 days with the biggest drops in revenue, 9 of them appeared on the 28th of the month. This is a very unique trend, and it is important for the coffee shops to further investigate why this day of the month is consistently underperforming. To counter this decline, the business could

implement targeted promotions such as end-of-month loyalty rewards, limited-time drink releases, '28th-only' discounts, or bundled offers to incentivize purchase.

3.6 Regression Analysis: Product Categories and Time-of-Day

To quantify which factors most strongly drive daily revenue, we constructed a multiple linear regression using daily units sold of each product category and revenue contributions from morning, afternoon, and evening time blocks. This model achieved an exceptionally high explanatory power of $R^2 = 0.983$, indicating that the combination of product categories and time-of-day revenue patterns explained nearly all variation in total daily revenue.

The regression revealed several important insights. Coffee beans had the strongest positive impact on overall revenue, indicating that customers who purchase coffee beans are likely high-value customers or that beans have higher margins relative to other categories. Conversely, packaged chocolate had a negative coefficient ($-\$30.66$ per additional unit sold), suggesting a cannibalization effect where packaged chocolate sales displace more profitable products. Additionally, the model showed that evening sales contributed the most revenue, followed by morning sales, with afternoon sales contributing the least. This nuanced understanding of category-specific and time-specific revenue drivers helped shape our recommendations for product positioning, inventory prioritization, and promotional strategy.

4. Results of Regression Models

4.1 Regression for time passed, month, and day of week (predicting Revenue)

OLS Regression Results						
=====						
Dep. Variable:	Revenue	R-squared:	0.900			
Model:	OLS	Adj. R-squared:	0.893			
Method:	Least Squares	F-statistic:	125.6			
Date:	Mon, 08 Dec 2025	Prob (F-statistic):	3.07e-77			
Time:	02:15:11	Log-Likelihood:	-1329.6			
No. Observations:	181	AIC:	2685.			
Df Residuals:	168	BIC:	2727.			
Df Model:	12					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]
const	2462.4197	111.954	21.995	0.000	2241.402	2683.437
t	9.0087	3.324	2.711	0.007	2.447	15.570
DayOfWeek_Friday	31.2965	108.122	0.289	0.773	-182.156	244.749
DayOfWeek_Monday	88.3518	108.047	0.818	0.415	-124.953	301.657
DayOfWeek_Saturday	50.4327	109.088	0.462	0.644	-164.928	265.793
DayOfWeek_Thursday	17.0276	108.134	0.157	0.875	-196.448	230.503
DayOfWeek_Tuesday	-6.0931	108.081	-0.056	0.955	-219.464	207.278
DayOfWeek_Wednesday	17.0182	108.123	0.157	0.875	-196.436	230.473
Month_2	-181.1369	141.221	-1.283	0.201	-459.934	97.660
Month_3	22.4795	219.825	0.102	0.919	-411.496	456.455
Month_4	523.7750	313.870	1.669	0.097	-95.863	1143.413
Month_5	1339.3703	410.968	3.259	0.001	528.044	2150.696
Month_6	1559.1387	510.171	3.056	0.003	551.967	2566.310
=====						
Omnibus:	18.219	Durbin-Watson:	1.015			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	20.760			
Skew:	-0.743	Prob(JB):	3.10e-05			
Kurtosis:	3.740	Cond. No.	2.74e+03			

We added a constant for days passed, and then created dummy variables for month and day of the week. We then used daily revenue as the y variable and tried to predict it using the regression. By doing so, we were able to see what days of the week were most profitable. Monday was the best performing day and Tuesday was the worst. We hypothesized that Monday was the best due to people going back to work, and wanting a coffee or treat to motivate them for the work week. We recommend that the coffee shop post promotions and advertisements for Tuesdays in order to promote revenue on these days.

We also found that January and February performed the worst, May and June performed the best. Due to the constraint of only having 6 months of data, it is hard for us to know if this is because of seasonality trends, or a clear upward pattern in sales. Either way, the coffee shop should try and promote spending in the winter, through things such as having unique seasonal drinks.

4.2 Regression for Product Category and Time of Day (predicting Revenue)

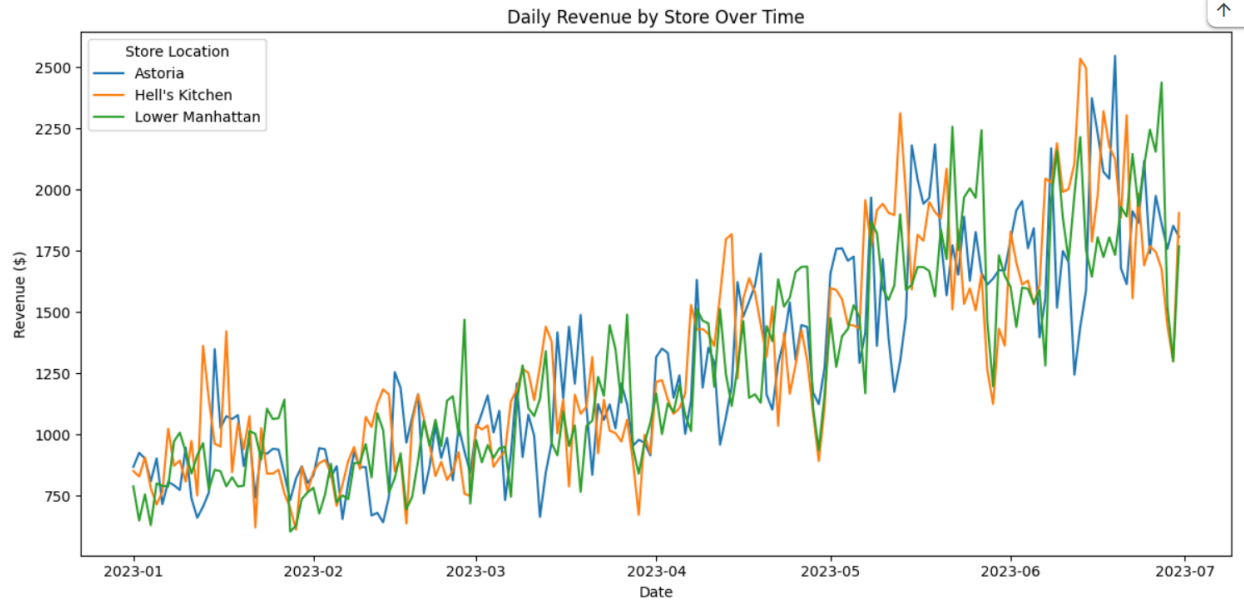
OLS Regression Results						
=====						
Dep. Variable:	Revenue	R-squared:	0.983			
Model:	OLS	Adj. R-squared:	0.982			
Method:	Least Squares	F-statistic:	966.7			
Date:	Mon, 08 Dec 2025	Prob (F-statistic):	3.56e-144			
Time:	02:15:32	Log-Likelihood:	-1170.5			
No. Observations:	181	AIC:	2363.			
Df Residuals:	170	BIC:	2398.			
Df Model:	10					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	13.7108	42.173	0.325	0.745	-69.539	96.960
TimeOfDay_Evening	2.5901	1.004	2.579	0.011	0.608	4.572
TimeOfDay_Morning	2.0246	0.481	4.206	0.000	1.074	2.975
product_category_Brande	16.8161	4.391	3.829	0.000	8.148	25.485
product_category_Coffee	4.5608	0.744	6.130	0.000	3.092	6.030
product_category_Coffee beans	18.1066	4.005	4.521	0.000	10.200	26.013
product_category_Drinking Chocolate	4.0021	1.675	2.389	0.018	0.695	7.309
product_category_Flavours	0.6722	1.548	0.434	0.665	-2.384	3.729
product_category_Loose Tea	0.3201	4.566	0.070	0.944	-8.694	9.334
product_category_Packaged Chocolate	-30.6559	8.565	-3.579	0.000	-47.564	-13.748
product_category_Tea	2.7452	0.780	3.519	0.001	1.205	4.285

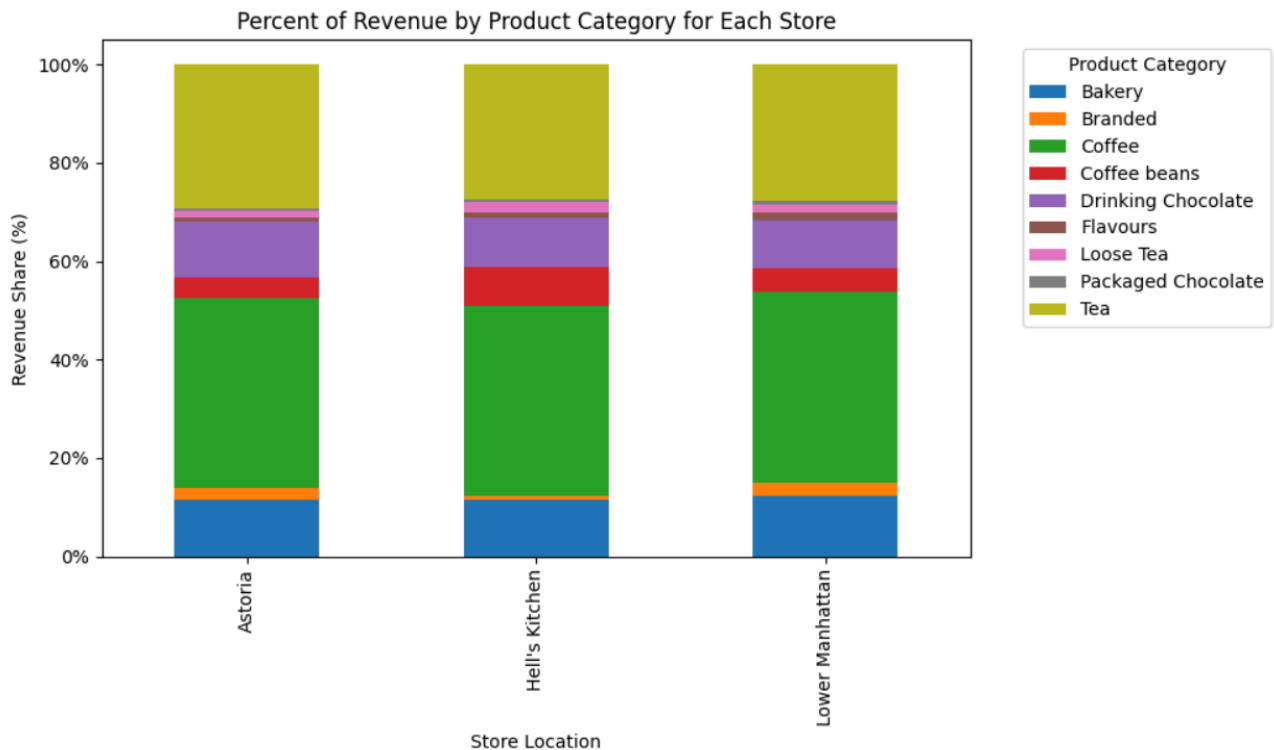
We created dummy variables for product categories (with bakery excluded) and time of day(afternoon excluded). We did this to see what coefficient was the greatest and which most positively affected revenue from a transaction. By creating dummy variables of 1 and 0, this allows us to compare all the coefficients of equal scale.

4.3 Visualizations that didn't lead to any feasible insights

We tried a few visualizations that didn't lead to too many patterns or insights. While not included in our presentation, it was still part of our process for understanding the data.



The first visualization showed daily revenue by each store over time. It is hard to clearly compare them and draw patterns, so we left this visualization out of our presentation.



This visualization shows each product categories percent of revenue at each store. They all looked very similar, and it was hard to draw any insights from this representation.

5. Limitations

While the dataset is extensive, several limitations should be acknowledged. First, the dataset covers only the year 2023, limiting our ability to identify long-term cyclical or multi-year seasonal trends. Second, the absence of customer demographic data prevents segmentation analysis or targeting by consumer profile. Third, the regression models face potential overfitting due to the inclusion of numerous predictors, some of which (e.g., weekday indicators) were statistically insignificant. Fourth, external factors such as weather conditions, holidays, local events, and competitor activity—which could meaningfully influence café sales—were not included in the dataset. Lastly, the data represents only three NYC locations, which may not generalize to other markets.

5. References

1. AhmedAbbas757. (2021). *Coffee Shop Sales* [Dataset]. Kaggle. <https://www.kaggle.com/datasets/ahmedabbas757/coffee-sales>